

Adroit Technologies

Protocol Driver

Name	Conet PCI Card API
DLL	CONETPCI



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Contents

1. Overview	4
2. Wiring	5
3. Device Configuration	6
3.1. Hardware Installation	6
3.2. Driver Installation and Setup.....	8
3.3. Device Configuration	9
3.3.1. Select FEP Card for this Device.....	9
3.3.2. FEP Card State	9
3.3.3. Outstation Settings	9
4. Supported Addresses	10
5. Protocol Definition and Considerations	11
5.1. PLC Software	11
6. General Notes and Troubleshooting	12
6.1. Contact Details.....	12
6.2. Adroit Technologies Knowledge Base.....	12
6.3. Troubleshooting	12
6.3.1. Communications problems.....	12
6.3.2. Installation problems.....	12
7. Appendix A – Node Address Table	13
8. Revisions.....	15
8.1. Document Revision.....	15
8.2. DLL Revision	15

1. Overview

The Conet driver makes use of third-party software to interface with the hardware. Therefore, to run this driver successfully, the "C6193A CONET PCI Interface Card" software must be installed. A Conet card is not required to configure. Follow the instructions in [Section 4. Device Configuration](#) for more details.

2. Wiring

The Conet network itself, including the wiring between the card and the stations, consists of a two-wire cable (usually shielded). As physical wiring is not within the scope of this document, please consult documentation accompanying the Conet PCI card for details.

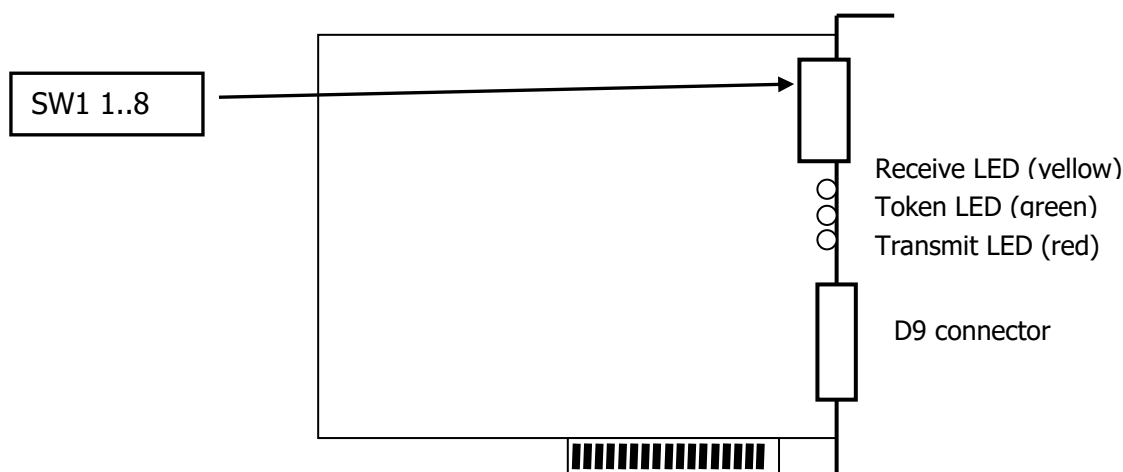
3. Device Configuration

3.1. Hardware Installation

Please follow the instructions provided with your card, the instructions below are correct at time of compilation, and may differ from the prescribed procedure.

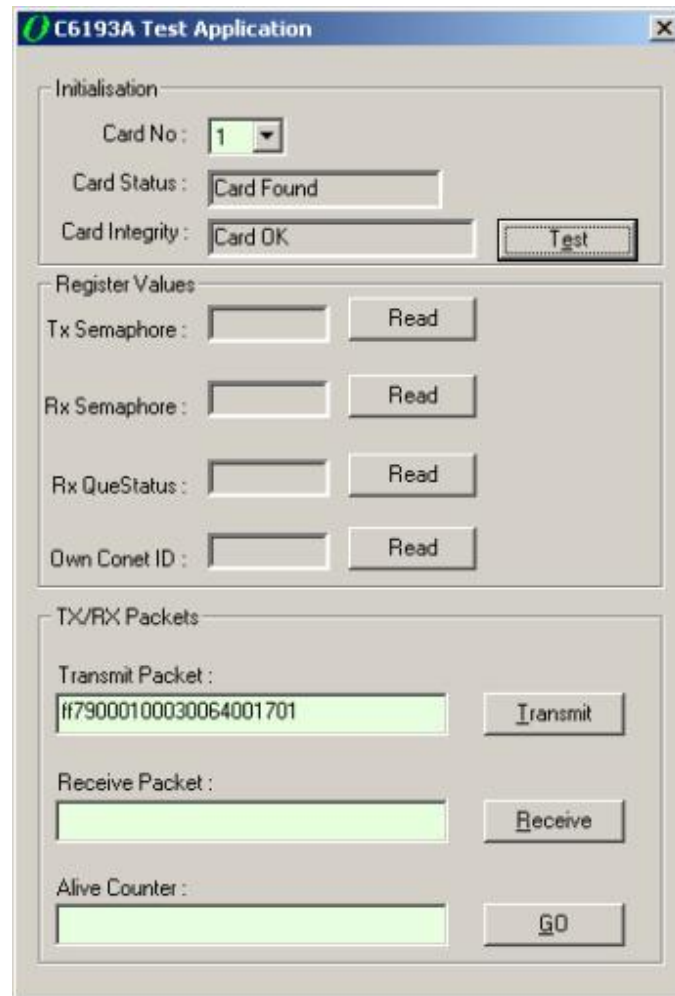
Unless the card is already successfully installed and working in the host machine the following steps must be followed.

1. Enable Plug-N-Play on your computer's BIOS.
2. Switch off your computer and unplug the power cable.
3. Insert the card into a blank PCI (white) slot.



4. Card OID /Node Address(SW1 1to SW1 7). The own ID of the card must be set to a unique address between 1 and 127. Remember that optimum performance (owing to the nature of the "Next Node searching algorithm employed for token passing) is achieved when there are no gaps between the node numbers.
5. Mode Setting (SW1 8). Select between (N)ormal mode and (S)low mode. Mode N is 62500 baud and Mode S is 7800 baud. Typically Mode N is used, but for networks with high interference levels, or large distances between the terminators, Mode S will be more reliable.
6. Turn on the computer, windows will automatically install the Jungo card drivers for you from the Conet Software CD, which must be inserted into your CD-Rom drive.
7. Run the Omniflex CD setup program to install the COM component needed by the Adroit driver, a small test program will also be installed at this time.

8. Run the test application to verify that the card is working.



9. This is the view from device manager, showing the Jungo driver installed correctly.



PLEASE NOTE

See table at end of this document for dip-switch positions.

If installation fails or the card does not work, please obtain updated device drivers directly from Omniflex as Adroit Technologies does not keep this software.

3.2. Driver Installation and Setup

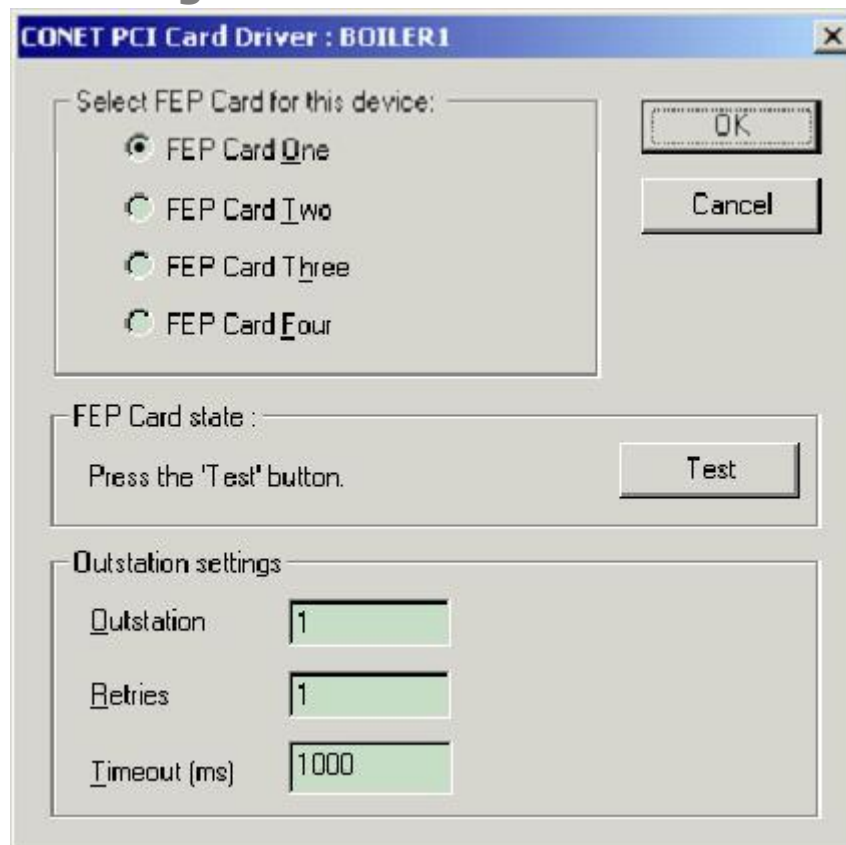
The driver installation follows the normal installation procedure. This will entail running Adroit setup and selecting the "Drivers" tab, choosing the "Add Driver" button, selecting the Conet PCI driver, and entering in the setup details in the setup dialogue that appears. However, because third-party software is used to assist in the low-level communications, there are certain checks that can be made to ensure that the installation was successful.

Perform the checks listed below if you have any problems.

Security Problems you may have:

Typically, when installing Adroit and configuring the driver for the first time, the configuring user will have Administrator Privileges allocated.

3.3. Device Configuration



The driver setup dialog will allow you to configure your driver devices. Configure one device per Conet Network Node. A maximum of 127 devices may be used at one time. More than 32 devices on the Conet Network may significantly reduce the speed of data transmission and is not recommended.

3.3.1. Select FEP Card for this Device

FEP Card (1 – 4) - This Radio Button Group requires the selection of a Card Number. This Card Number has nothing to do with settings on the proprietary card itself; the Card Number is a driver option, which will be allocated to a physical card to identify it. This means that a maximum of four Conet cards may be installed in a single machine.

3.3.2. FEP Card State

TEST - Test button to verify that the device driver and card are functioning.

3.3.3. Outstation Settings

Outstation - (1-127) The outstation number is the Own ID (OID) of the device on the Conet Network with which this device must communicate.

Retries - Before failing a communications transaction, the device has the option of retrying. The value here will be the number of times the device will retry.

Timeout (ms) - In version 6c of the driver onwards, the retries setting does now work, and the dialog appears differently. The new offline retry time however does not have any effect.

4. Supported Addresses

The format of the address string for normal communications is as follows:

DIT xxxx[.yy]

Where:

- DIT is a prefix or mnemonic for Data Interchange Table (i.e. the register table)
- 'xxxx' is the byte offset into the DIT table. Optionally a bit offset '[.yy]' may appear for Boolean scanning.

Device	Max Range	Boolean	Integer	Real
Digital	DIT 99999.15	DIT 100.12		
Analogue	DIT 99999		DIT 123	DIT 123



PLEASE NOTE

If after selecting a device, you do not see a default address appear in the Source Address field of the scanning dialog, the driver has not loaded. Look at the event viewer to establish the reason.

5. Protocol Definition and Considerations

The Conet PCI message protocol is shown in the HEX Datascope diagnostic displays as follows:

Read Request

DID	SID	Count	Res.	CMD	StartReg	Range
00 02	00 01	00 03	00	64	00 00	01

Read Response

DID	SID	Count	Res.	CMD	StartReg	Data
00 01	00 02	00 03	00	65	00 00	01 06

Write Request

device	Bytes	Command	StartReg	Range	Data
01	05	66	00 00	01	00 7F

Write Response

device	bytes	Command	StartReg
07	00	xx	Xx xx

device = (OID) of the targeted device on the Conet network.

Bytes = number of bytes in the telegram, excluding the first three.

Command code: 64h = 100 etc.

StartReg = offset into the DIT table from which data is to be read or to which data must be written.

Range = number of consecutive bytes that must be read or written from/to the DIT table.

Data = actual data returned by the device or written to the device.

5.1. PLC Software

No additional software is required, install the windows device driver for the PCI card from the CD supplied with the card.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

6.3.1. Communications problems

Make sure that once connected up, the card keeps passing the token. I.e. the green led should flash steadily. If it stops for a "breath" you may notice that this coincides with a communications error, check wiring or replace the interface card.

6.3.2. Installation problems

See the section "Driver installation and setup" for the correct installation procedures, and a list of checks to ensure that the device is correctly loaded.

7. Appendix A – Node Address Table

SW 1	SW 2	SW 3	SW 4	SW 5	SW 6	SW 7	Node #
ON	ON	ON	ON	ON	ON	ON	Invalid
ON	ON	ON	ON	ON	ON	OFF	1
ON	ON	ON	ON	ON	OFF	ON	2
ON	ON	ON	ON	ON	OFF	OFF	3
ON	ON	ON	ON	OFF	ON	ON	4
ON	ON	ON	ON	OFF	ON	OFF	5
ON	ON	ON	ON	OFF	OFF	ON	6
ON	ON	ON	ON	OFF	OFF	OFF	7
ON	ON	ON	OFF	ON	ON	ON	8
ON	ON	ON	OFF	ON	ON	OFF	9
ON	ON	ON	OFF	ON	OFF	ON	10
ON	ON	ON	OFF	ON	OFF	OFF	11
ON	ON	ON	OFF	OFF	ON	ON	12
ON	ON	ON	OFF	OFF	ON	OFF	13
ON	ON	ON	OFF	OFF	OFF	ON	14
ON	ON	ON	OFF	OFF	OFF	OFF	15
ON	ON	OFF	ON	ON	ON	ON	16
ON	ON	OFF	ON	ON	ON	OFF	17
ON	ON	OFF	ON	ON	OFF	ON	18
ON	ON	OFF	ON	ON	OFF	OFF	19
ON	ON	OFF	ON	OFF	ON	ON	20
ON	ON	OFF	ON	OFF	ON	OFF	21
ON	ON	OFF	ON	OFF	OFF	ON	22
ON	ON	OFF	ON	OFF	OFF	OFF	23
ON	ON	OFF	OFF	ON	ON	ON	24
ON	ON	OFF	OFF	ON	ON	OFF	25
ON	ON	OFF	OFF	ON	OFF	ON	26
ON	ON	OFF	OFF	ON	OFF	OFF	27
ON	ON	OFF	OFF	OFF	ON	ON	28
ON	ON	OFF	OFF	OFF	ON	OFF	29
ON	ON	OFF	OFF	OFF	OFF	ON	30
ON	ON	OFF	OFF	OFF	OFF	OFF	31
ON	OFF	ON	ON	ON	ON	ON	32
ON	OFF	ON	ON	ON	ON	OFF	33
ON	OFF	ON	ON	ON	OFF	ON	34
ON	OFF	ON	ON	ON	OFF	OFF	35
ON	OFF	ON	ON	OFF	ON	ON	36
ON	OFF	ON	ON	OFF	ON	OFF	37
ON	OFF	ON	ON	OFF	OFF	ON	38
ON	OFF	ON	ON	OFF	OFF	OFF	39
ON	OFF	ON	OFF	ON	ON	ON	40
ON	OFF	ON	OFF	ON	ON	OFF	41
ON	OFF	ON	OFF	ON	OFF	ON	42
ON	OFF	ON	OFF	ON	OFF	OFF	43
ON	OFF	ON	OFF	OFF	ON	ON	44
ON	OFF	ON	OFF	OFF	ON	OFF	45
ON	OFF	ON	OFF	OFF	OFF	ON	46
ON	OFF	ON	OFF	OFF	OFF	OFF	47
ON	OFF	OFF	ON	ON	ON	ON	48

SW 1	SW 2	SW 3	SW 4	SW 5	SW 6	SW 7	Node #
ON	OFF	OFF	ON	ON	ON	OFF	49
ON	OFF	OFF	ON	ON	OFF	ON	50
ON	OFF	OFF	ON	ON	OFF	OFF	51
ON	OFF	OFF	ON	OFF	ON	ON	52
ON	OFF	OFF	ON	OFF	ON	OFF	53
ON	OFF	OFF	ON	OFF	OFF	ON	54
ON	OFF	OFF	ON	OFF	OFF	OFF	55
ON	OFF	OFF	OFF	ON	ON	ON	56
ON	OFF	OFF	OFF	ON	ON	OFF	57
ON	OFF	OFF	OFF	ON	OFF	ON	58
ON	ON	OFF	OFF	ON	OFF	OFF	59
ON	ON	OFF	OFF	OFF	ON	ON	60
ON	ON	OFF	OFF	OFF	ON	OFF	61
ON	ON	OFF	OFF	OFF	OFF	ON	62
ON	ON	OFF	OFF	OFF	OFF	OFF	63
ON	OFF	ON	ON	ON	ON	ON	64
ON	OFF	ON	ON	ON	ON	OFF	65
ON	OFF	ON	ON	ON	OFF	ON	66
ON	OFF	ON	ON	ON	OFF	OFF	67
ON	OFF	ON	ON	OFF	ON	ON	68
ON	OFF	ON	ON	OFF	ON	OFF	69
ON	OFF	ON	ON	OFF	OFF	ON	70
ON	OFF	ON	ON	OFF	OFF	OFF	71
ON	OFF	ON	OFF	ON	ON	ON	72
ON	OFF	ON	OFF	ON	ON	OFF	73
ON	OFF	ON	OFF	ON	OFF	ON	74
ON	OFF	ON	OFF	ON	OFF	OFF	75
ON	OFF	ON	OFF	OFF	ON	ON	76
ON	OFF	ON	OFF	OFF	ON	OFF	77
ON	OFF	ON	OFF	OFF	OFF	ON	78
ON	OFF	ON	OFF	OFF	OFF	OFF	79
ON	OFF	OFF	ON	ON	ON	ON	80
ON	OFF	OFF	ON	ON	ON	OFF	81
ON	OFF	OFF	ON	ON	OFF	ON	82
ON	OFF	OFF	ON	ON	OFF	OFF	83
ON	OFF	OFF	ON	OFF	ON	ON	84
ON	OFF	OFF	ON	OFF	ON	OFF	85
ON	OFF	OFF	ON	OFF	OFF	ON	86
ON	OFF	OFF	ON	OFF	OFF	OFF	87
ON	OFF	OFF	OFF	ON	ON	ON	88
ON	OFF	OFF	OFF	ON	ON	OFF	89
ON	OFF	OFF	OFF	ON	OFF	ON	90
ON	OFF	OFF	OFF	ON	OFF	OFF	91
ON	OFF	OFF	OFF	OFF	ON	ON	92
ON	OFF	OFF	OFF	OFF	ON	OFF	93
ON	OFF	OFF	OFF	OFF	OFF	ON	94
ON	OFF	OFF	OFF	OFF	OFF	OFF	95
OFF	ON	ON	ON	ON	ON	ON	96
OFF	ON	ON	ON	ON	ON	OFF	97
OFF	ON	ON	ON	ON	OFF	ON	98
OFF	ON	ON	ON	ON	OFF	OFF	99
OFF	ON	ON	ON	OFF	ON	ON	100
OFF	ON	ON	ON	OFF	ON	OFF	101
OFF	ON	ON	ON	OFF	OFF	ON	102
OFF	ON	ON	ON	OFF	OFF	OFF	103
OFF	ON	ON	OFF	ON	ON	ON	104
OFF	ON	ON	OFF	ON	ON	OFF	105

SW 1	SW 2	SW 3	SW 4	SW 5	SW 6	SW 7	Node #
OFF	ON	ON	OFF	ON	OFF	ON	106
OFF	ON	ON	OFF	ON	OFF	OFF	107
OFF	ON	ON	OFF	OFF	ON	ON	108
OFF	ON	ON	OFF	OFF	ON	OFF	109
OFF	ON	ON	OFF	OFF	OFF	ON	110
OFF	ON	ON	OFF	OFF	OFF	OFF	111
OFF	ON	OFF	ON	ON	ON	ON	112
OFF	ON	OFF	ON	ON	ON	OFF	113
OFF	ON	OFF	ON	ON	OFF	ON	114
ON	OFF	OFF	ON	ON	OFF	OFF	115
ON	OFF	OFF	ON	OFF	ON	ON	116
ON	OFF	OFF	ON	OFF	ON	OFF	117
ON	OFF	OFF	ON	OFF	OFF	ON	118
ON	OFF	OFF	ON	OFF	OFF	OFF	119
ON	OFF	OFF	OFF	ON	ON	ON	120
ON	OFF	OFF	OFF	ON	ON	OFF	121
ON	OFF	OFF	OFF	ON	OFF	ON	122
ON	ON	OFF	OFF	ON	OFF	OFF	123
ON	ON	OFF	OFF	OFF	ON	ON	124
ON	ON	OFF	OFF	OFF	ON	OFF	125
ON	ON	OFF	OFF	OFF	OFF	ON	126
ON	ON	OFF	OFF	OFF	OFF	OFF	127

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan/2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				

8.2. DLL Revision

Rev.	Date	Changes
1	24-Apr-1998	First complete revision (ISA CARD)
	7-Mar-2003	PCI card version documented correctly.
	15-Nov-2004	Retries works in 6c, card-installation instruction expanded.
8	28-Spe-2018	Performance counts did not work when running the Agent Server as a service.